**BSc (Hons) in Information Technology**

**Year 4 Semester 2**

Lab Exercise 04 – Resource Standard Metric

Group Details

| Student Name | IT Number |
|---|---|
| Bhanuka K. Y | IT22563750 |
| Eranga N.G.A.H | IT22604958 |
| Hewamanna K. T | IT22584090 |
| Vidurangi K.G. D | IT22258694 |

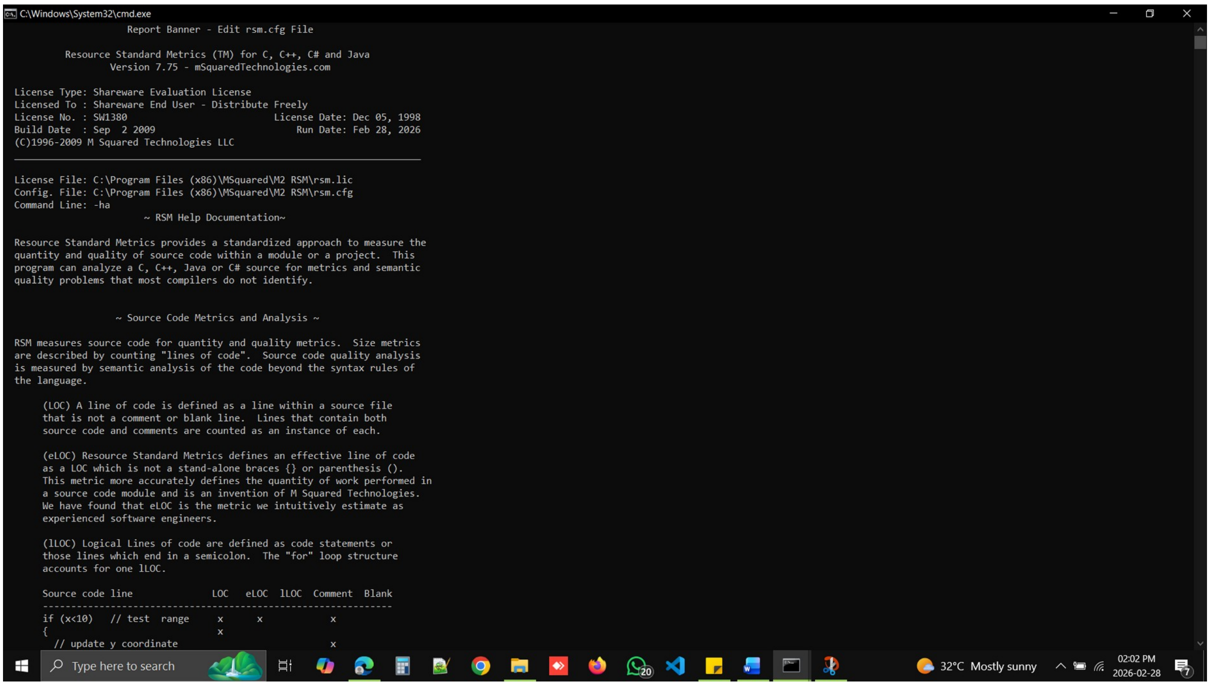
# Table of Contents

# RSM – Resource Standard Matrix

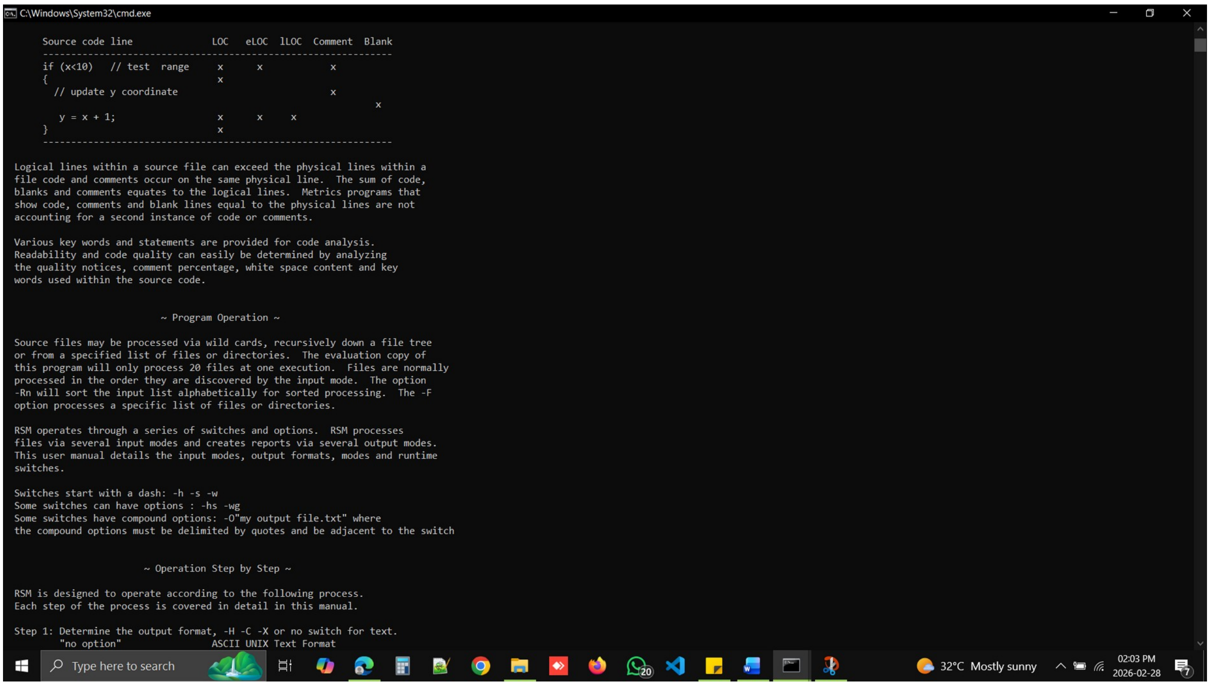
RSM (Resource Standard Matrix) is a tool used in the management of projects and in software engineering to identify the necessary resources to be used in the execution of the activities involved in the project.

## View the complete RSM documentation type ---- rsm –ha

```
C:\Users\Sajani Hewamanne\Desktop\Y4S2\SQA\Lab Sheet\Lab Sheet 4\Lab 4\Java Files>rsm -ha
```

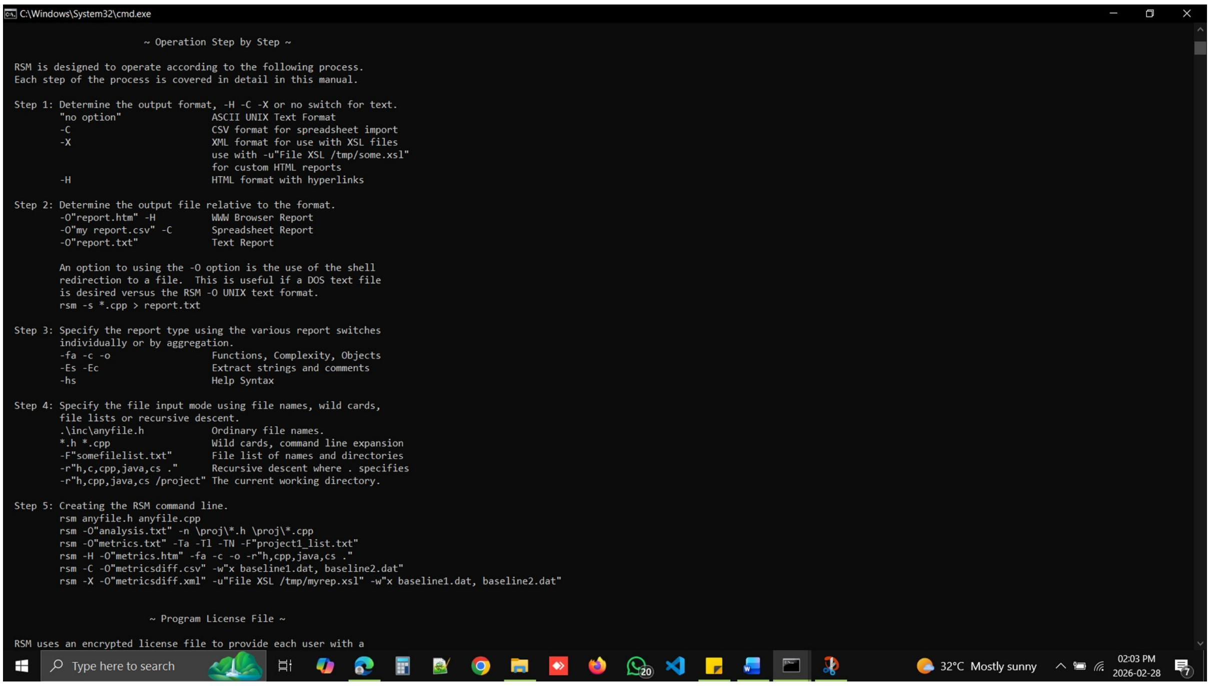


```
C:\Windows\System32\cmd.exe
                      Report Banner - Edit rsm.cfg File

           Resource Standard Metrics (TM) for C, C++, C# and Java
                   Version 7.75 - mSquaredTechnologies.com

License Type: Shareware Evaluation License
Licensed To : Shareware End User - Distribute Freely
License No. : SW1380                          License Date: Dec 05, 1998
Build Date  : Sep  2 2009                         Run Date: Feb 28, 2026
(C)1996-2009 M Squared Technologies LLC
________________________________________________________________________

License File: C:\Program Files (x86)\MSquared\M2 RSM\rsm.lic
Config. File: C:\Program Files (x86)\MSquared\M2 RSM\rsm.cfg
Command Line: -ha
                       ~ RSM Help Documentation~

Resource Standard Metrics provides a standardized approach to measure the
quantity and quality of source code within a module or a project.  This
program can analyze a C, C++, Java or C# source for metrics and semantic
quality problems that most compilers do not identify.


                  ~ Source Code Metrics and Analysis ~

RSM measures source code for quantity and quality metrics.  Size metrics
are described by counting "lines of code".  Source code quality analysis
is measured by semantic analysis of the code beyond the syntax rules of
the language.

     (LOC) A line of code is defined as a line within a source file
     that is not a comment or blank line.  Lines that contain both
     source code and comments are counted as an instance of each.

     (eLOC) Resource Standard Metrics defines an effective line of code
     as a LOC which is not a stand-alone braces {} or parenthesis ().
     This metric more accurately defines the quantity of work performed in
     a source code module and is an invention of M Squared Technologies.
     We have found that eLOC is the metric we intuitively estimate as
     experienced software engineers.

     (lLOC) Logical Lines of code are defined as code statements or
     those lines which end in a semicolon.  The "for" loop structure
     accounts for one lLOC.

     Source code line              LOC   eLOC  lLOC  Comment  Blank
     --------------------------------------------------------------
     if (x<10)   // test  range     x      x            x
     {                              x
       // update y coordinate                           x
```

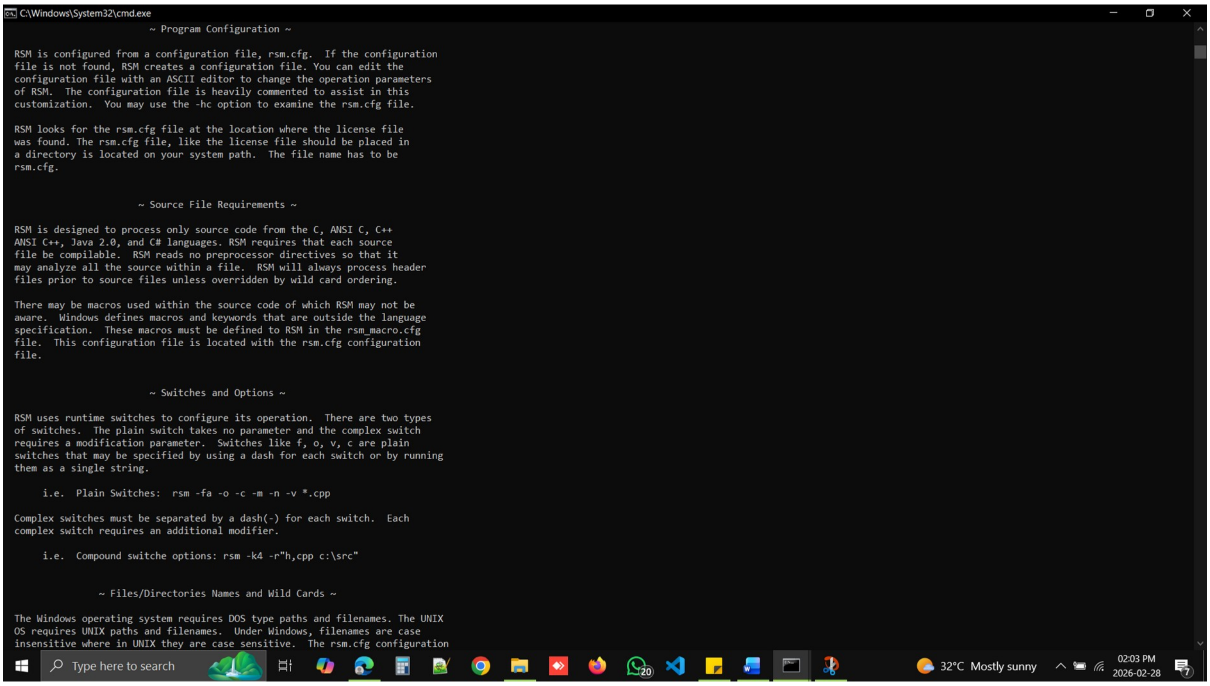


```
C:\Windows\System32\cmd.exe

     Source code line              LOC   eLOC  lLOC  Comment  Blank
     ---------------------------------------------------------------
     if (x<10)   // test  range     x      x           x
     {                              x
       // update y coordinate                          x
                                                              x
        y = x + 1;                  x      x     x
     }                              x
     ---------------------------------------------------------------

Logical lines within a source file can exceed the physical lines within a
file code and comments occur on the same physical line.  The sum of code,
blanks and comments equates to the logical lines.  Metrics programs that
show code, comments and blank lines equal to the physical lines are not
accounting for a second instance of code or comments.

Various key words and statements are provided for code analysis.
Readability and code quality can easily be determined by analyzing
the quality notices, comment percentage, white space content and key
words used within the source code.


                          ~ Program Operation ~

Source files may be processed via wild cards, recursively down a file tree
or from a specified list of files or directories.  The evaluation copy of
this program will only process 20 files at one execution.  Files are normally
processed in the order they are discovered by the input mode.  The option
-Rn will sort the input list alphabetically for sorted processing.  The -F
option processes a specific list of files or directories.

RSM operates through a series of switches and options.  RSM processes
files via several input modes and creates reports via several output modes.
This user manual details the input modes, output formats, modes and runtime
switches.

Switches start with a dash: -h -s -w
Some switches can have options : -hs -wg
Some switches have compound options: -O"my output file.txt" where
the compound options must be delimited by quotes and be adjacent to the switch


                       ~ Operation Step by Step ~

RSM is designed to operate according to the following process.
Each step of the process is covered in detail in this manual.

Step 1: Determine the output format, -H -C -X or no switch for text.
        "no option"                ASCII UNIX Text Format
```

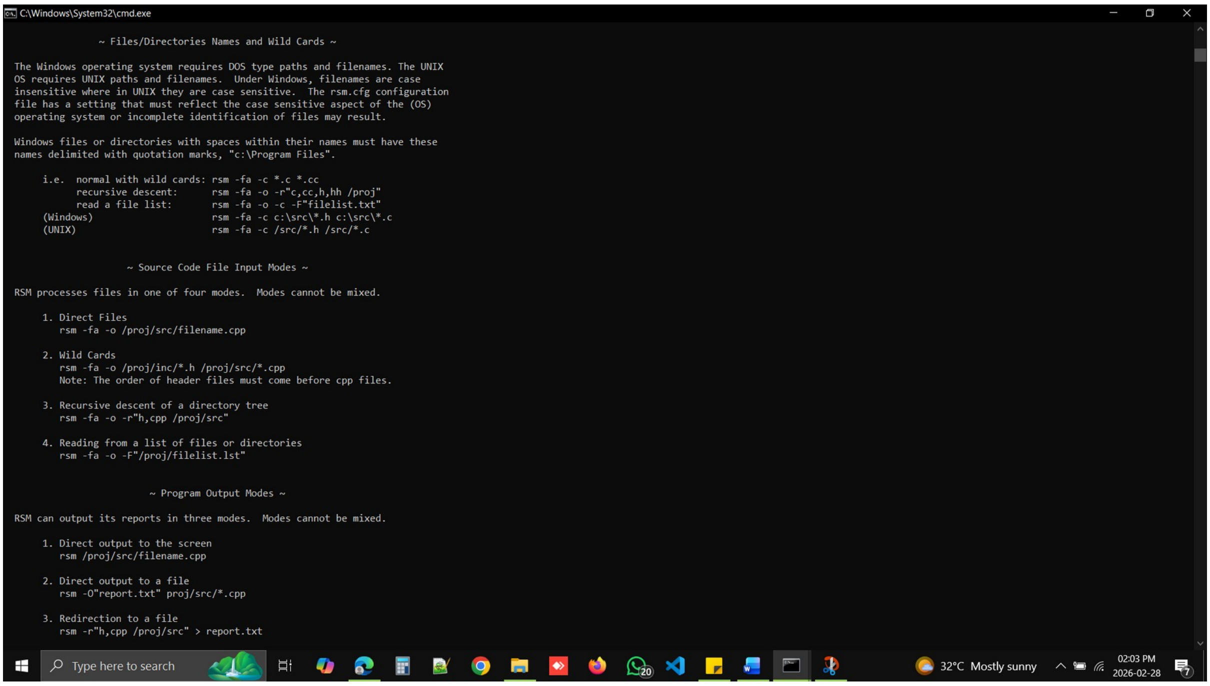




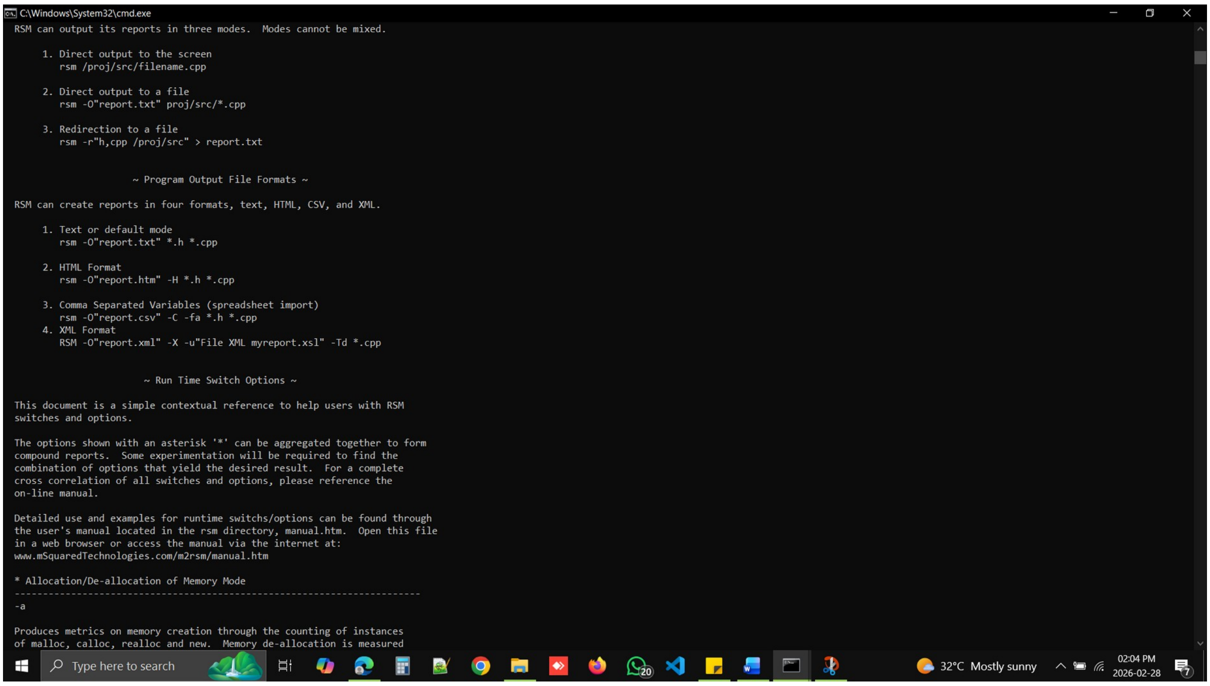
```
C:\Windows\System32\cmd.exe

                       ~ Operation Step by Step ~

RSM is designed to operate according to the following process.
Each step of the process is covered in detail in this manual.

Step 1: Determine the output format, -H -C -X or no switch for text.
        "no option"                ASCII UNIX Text Format
        -C                         CSV format for spreadsheet import
        -X                         XML format for use with XSL files
                                   use with -u"File XSL /tmp/some.xsl"
                                   for custom HTML reports
        -H                         HTML format with hyperlinks

Step 2: Determine the output file relative to the format.
        -O"report.htm" -H          WWW Browser Report
        -O"my report.csv" -C       Spreadsheet Report
        -O"report.txt"             Text Report

        An option to using the -O option is the use of the shell
        redirection to a file.  This is useful if a DOS text file
        is desired versus the RSM -O UNIX text format.
        rsm -s *.cpp > report.txt

Step 3: Specify the report type using the various report switches
        individually or by aggregation.
        -fa -c -o                  Functions, Complexity, Objects
        -Es -Ec                    Extract strings and comments
        -hs                        Help Syntax

Step 4: Specify the file input mode using file names, wild cards,
        file lists or recursive descent.
        .\inc\anyfile.h            Ordinary file names.
        *.h *.cpp                  Wild cards, command line expansion
        -F"somefilelist.txt"       File list of names and directories
        -r"h,c,cpp,java,cs ."      Recursive descent where . specifies
        -r"h,cpp,java,cs /project" The current working directory.

Step 5: Creating the RSM command line.
        rsm anyfile.h anyfile.cpp
        rsm -O"analysis.txt" -n \proj\*.h \proj\*.cpp
        rsm -O"metrics.txt" -Ta -Tl -TN -F"project1_list.txt"
        rsm -H -O"metrics.htm" -fa -c -o -r"h,cpp,java,cs ."
        rsm -C -O"metricsdiff.csv" -w"x baseline1.dat, baseline2.dat"
        rsm -X -O"metricsdiff.xml" -u"File XSL /tmp/myrep.xsl" -w"x baseline1.dat, baseline2.dat"


                        ~ Program License File ~

RSM uses an encrypted license file to provide each user with a
```



C:\Windows\System32\cmd.exe
~ Program Configuration ~
RSM is configured from a configuration file, rsm.cfg. If the configuration file is not found, RSM creates a configuration file. You can edit the configuration file with an ASCII editor to change the operation parameters of RSM. The configuration file is heavily commented to assist in this customization. You may use the -hc option to examine the rsm.cfg file.
RSM looks for the rsm.cfg file at the location where the license file was found. The rsm.cfg file, like the license file should be placed in a directory is located on your system path. The file name has to be rsm.cfg.
~ Source File Requirements ~
RSM is designed to process only source code from the C, ANSI C, C++ ANSI C++, Java 2.0, and C# languages. RSM requires that each source file be compilable. RSM reads no preprocessor directives so that it may analyze all the source within a file. RSM will always process header files prior to source files unless overridden by wild card ordering.
There may be macros used within the source code of which RSM may not be aware. Windows defines macros and keywords that are outside the language specification. These macros must be defined to RSM in the rsm_macro.cfg file. This configuration file is located with the rsm.cfg configuration file.
~ Switches and Options ~
RSM uses runtime switches to configure its operation. There are two types of switches. The plain switch takes no parameter and the complex switch requires a modification parameter. Switches like f, o, v, c are plain switches that may be specified by using a dash for each switch or by running them as a single string.
i.e. Plain Switches: rsm -fa -o -c -m -n -v *.cpp
Complex switches must be separated by a dash(-) for each switch. Each complex switch requires an additional modifier.
i.e. Compound switche options: rsm -k4 -r"h,cpp c:\src"
~ Files/Directories Names and Wild Cards ~
The Windows operating system requires DOS type paths and filenames. The UNIX OS requires UNIX paths and filenames. Under Windows, filenames are case insensitive where in UNIX they are case sensitive. The rsm.cfg configuration
Type here to search
32°C Mostly sunny
02:03 PM
2026-02-28



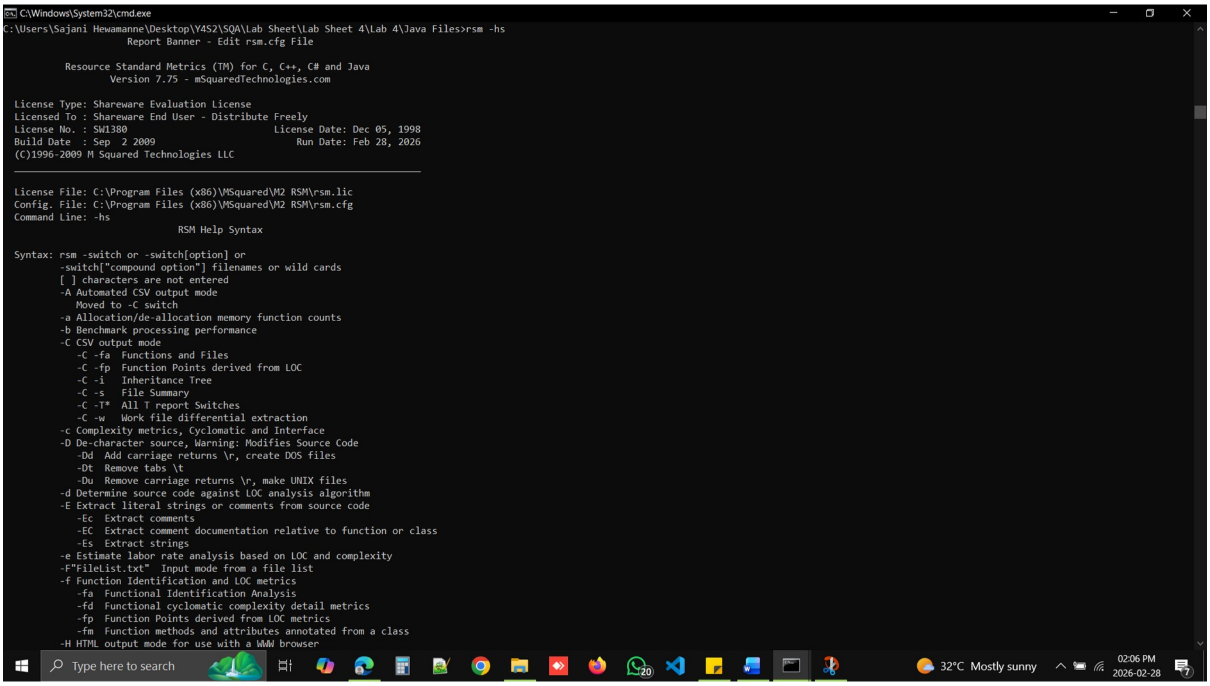
```
C:\Windows\System32\cmd.exe

              ~ Files/Directories Names and Wild Cards ~

The Windows operating system requires DOS type paths and filenames. The UNIX
OS requires UNIX paths and filenames.  Under Windows, filenames are case
insensitive where in UNIX they are case sensitive.  The rsm.cfg configuration
file has a setting that must reflect the case sensitive aspect of the (OS)
operating system or incomplete identification of files may result.

Windows files or directories with spaces within their names must have these
names delimited with quotation marks, "c:\Program Files".

     i.e.  normal with wild cards: rsm -fa -c *.c *.cc
           recursive descent:      rsm -fa -o -r"c,cc,h,hh /proj"
           read a file list:       rsm -fa -o -c -F"filelist.txt"
     (Windows)                     rsm -fa -c c:\src\*.h c:\src\*.c
     (UNIX)                        rsm -fa -c /src/*.h /src/*.c


                   ~ Source Code File Input Modes ~

RSM processes files in one of four modes.  Modes cannot be mixed.

     1. Direct Files
        rsm -fa -o /proj/src/filename.cpp

     2. Wild Cards
        rsm -fa -o /proj/inc/*.h /proj/src/*.cpp
        Note: The order of header files must come before cpp files.

     3. Recursive descent of a directory tree
        rsm -fa -o -r"h,cpp /proj/src"

     4. Reading from a list of files or directories
        rsm -fa -o -F"/proj/filelist.lst"


                       ~ Program Output Modes ~

RSM can output its reports in three modes.  Modes cannot be mixed.

     1. Direct output to the screen
        rsm /proj/src/filename.cpp

     2. Direct output to a file
        rsm -O"report.txt" proj/src/*.cpp

     3. Redirection to a file
        rsm -r"h,cpp /proj/src" > report.txt
```



```
C:\Windows\System32\cmd.exe
RSM can output its reports in three modes.  Modes cannot be mixed.

     1. Direct output to the screen
        rsm /proj/src/filename.cpp

     2. Direct output to a file
        rsm -O"report.txt" proj/src/*.cpp

     3. Redirection to a file
        rsm -r"h,cpp /proj/src" > report.txt


                    ~ Program Output File Formats ~

RSM can create reports in four formats, text, HTML, CSV, and XML.

     1. Text or default mode
        rsm -O"report.txt" *.h *.cpp

     2. HTML Format
        rsm -O"report.htm" -H *.h *.cpp

     3. Comma Separated Variables (spreadsheet import)
        rsm -O"report.csv" -C -fa *.h *.cpp
     4. XML Format
        RSM -O"report.xml" -X -u"File XML myreport.xsl" -Td *.cpp


                      ~ Run Time Switch Options ~

This document is a simple contextual reference to help users with RSM
switches and options.

The options shown with an asterisk '*' can be aggregated together to form
compound reports.  Some experimentation will be required to find the
combination of options that yield the desired result.  For a complete
cross correlation of all switches and options, please reference the
on-line manual.

Detailed use and examples for runtime switchs/options can be found through
the user's manual located in the rsm directory, manual.htm.  Open this file
in a web browser or access the manual via the internet at:
www.mSquaredTechnologies.com/m2rsm/manual.htm

* Allocation/De-allocation of Memory Mode
-------------------------------------------------------------------------
-a

Produces metrics on memory creation through the counting of instances
of malloc, calloc, realloc and new.  Memory de-allocation is measured
```

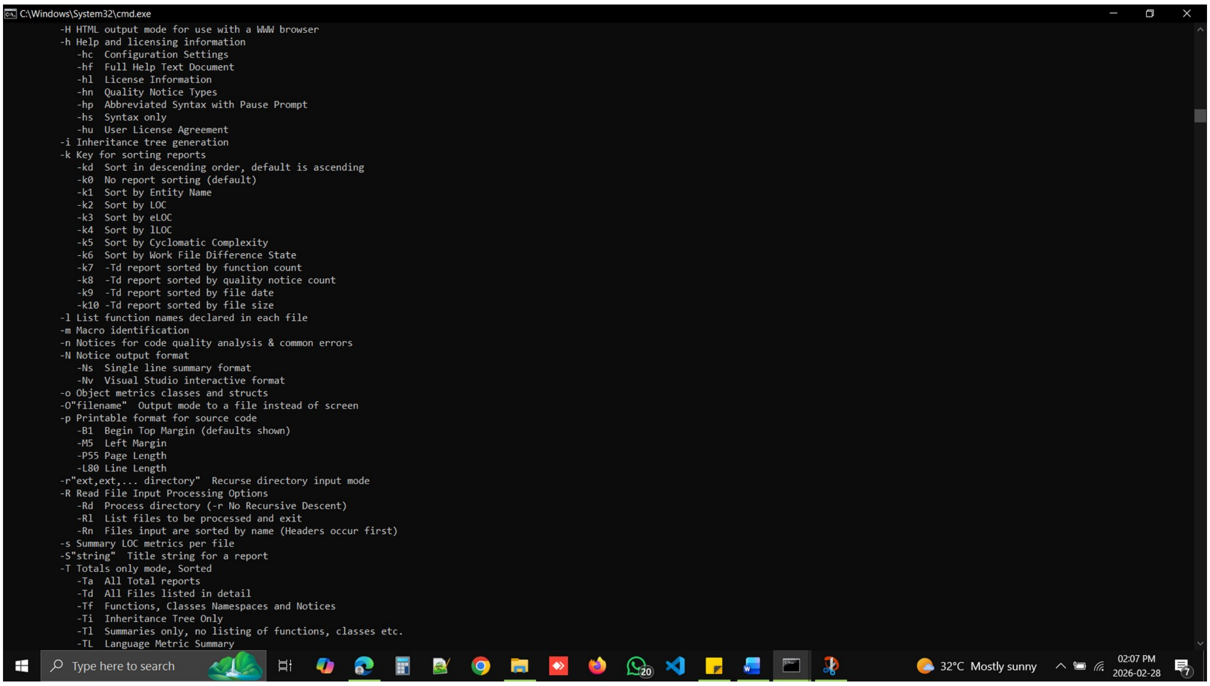
## View the basic commands and options type ----- rsm -hs

```
C:\Users\Sajani Hewamanne\Desktop\Y4S2\SQA\Lab Sheet\Lab Sheet 4\Lab 4\Java Files>rsm -hs
```



```
C:\Windows\System32\cmd.exe
C:\Users\Sajani Hewamanne\Desktop\Y4S2\SQA\Lab Sheet\Lab Sheet 4\Lab 4\Java Files>rsm -hs
                      Report Banner - Edit rsm.cfg File

           Resource Standard Metrics (TM) for C, C++, C# and Java
                   Version 7.75 - mSquaredTechnologies.com

  License Type: Shareware Evaluation License
  Licensed To : Shareware End User - Distribute Freely
  License No. : SW1380                          License Date: Dec 05, 1998
  Build Date  : Sep  2 2009                         Run Date: Feb 28, 2026
  (C)1996-2009 M Squared Technologies LLC
  ________________________________________________________________________

  License File: C:\Program Files (x86)\MSquared\M2 RSM\rsm.lic
  Config. File: C:\Program Files (x86)\MSquared\M2 RSM\rsm.cfg
  Command Line: -hs
                                RSM Help Syntax

  Syntax: rsm -switch or -switch[option] or
          -switch["compound option"] filenames or wild cards
          [ ] characters are not entered
          -A Automated CSV output mode
             Moved to -C switch
          -a Allocation/de-allocation memory function counts
          -b Benchmark processing performance
          -C CSV output mode
             -C -fa  Functions and Files
             -C -fp  Function Points derived from LOC
             -C -i   Inheritance Tree
             -C -s   File Summary
             -C -T*  All T report Switches
             -C -w   Work file differential extraction
          -c Complexity metrics, Cyclomatic and Interface
          -D De-character source, Warning: Modifies Source Code
             -Dd  Add carriage returns \r, create DOS files
             -Dt  Remove tabs \t
             -Du  Remove carriage returns \r, make UNIX files
          -d Determine source code against LOC analysis algorithm
          -E Extract literal strings or comments from source code
             -Ec  Extract comments
             -EC  Extract comment documentation relative to function or class
             -Es  Extract strings
          -e Estimate labor rate analysis based on LOC and complexity
          -F"FileList.txt"  Input mode from a file list
          -f Function Identification and LOC metrics
             -fa  Functional Identification Analysis
             -fd  Functional cyclomatic complexity detail metrics
             -fp  Function Points derived from LOC metrics
             -fm  Function methods and attributes annotated from a class
          -H HTML output mode for use with a WWW browser
```





```
C:\Windows\System32\cmd.exe
          -H HTML output mode for use with a WWW browser
          -h Help and licensing information
             -hc  Configuration Settings
             -hf  Full Help Text Document
             -hl  License Information
             -hn  Quality Notice Types
             -hp  Abbreviated Syntax with Pause Prompt
             -hs  Syntax only
             -hu  User License Agreement
          -i Inheritance tree generation
          -k Key for sorting reports
             -kd  Sort in descending order, default is ascending
             -k0  No report sorting (default)
             -k1  Sort by Entity Name
             -k2  Sort by LOC
             -k3  Sort by eLOC
             -k4  Sort by lLOC
             -k5  Sort by Cyclomatic Complexity
             -k6  Sort by Work File Difference State
             -k7  -Td report sorted by function count
             -k8  -Td report sorted by quality notice count
             -k9  -Td report sorted by file date
             -k10 -Td report sorted by file size
          -l List function names declared in each file
          -m Macro identification
          -n Notices for code quality analysis & common errors
          -N Notice output format
             -Ns  Single line summary format
             -Nv  Visual Studio interactive format
          -o Object metrics classes and structs
          -O"filename"  Output mode to a file instead of screen
          -p Printable format for source code
             -B1  Begin Top Margin (defaults shown)
             -M5  Left Margin
             -P55 Page Length
             -L80 Line Length
          -r"ext,ext,... directory"  Recurse directory input mode
          -R Read File Input Processing Options
             -Rd  Process directory (-r No Recursive Descent)
             -Rl  List files to be processed and exit
             -Rn  Files input are sorted by name (Headers occur first)
          -s Summary LOC metrics per file
          -S"string"  Title string for a report
          -T Totals only mode, Sorted
             -Ta  All Total reports
             -Td  All Files listed in detail
             -Tf  Functions, Classes Namespaces and Notices
             -Ti  Inheritance Tree Only
             -Tl  Summaries only, no listing of functions, classes etc.
             -TL  Language Metric Summary
```

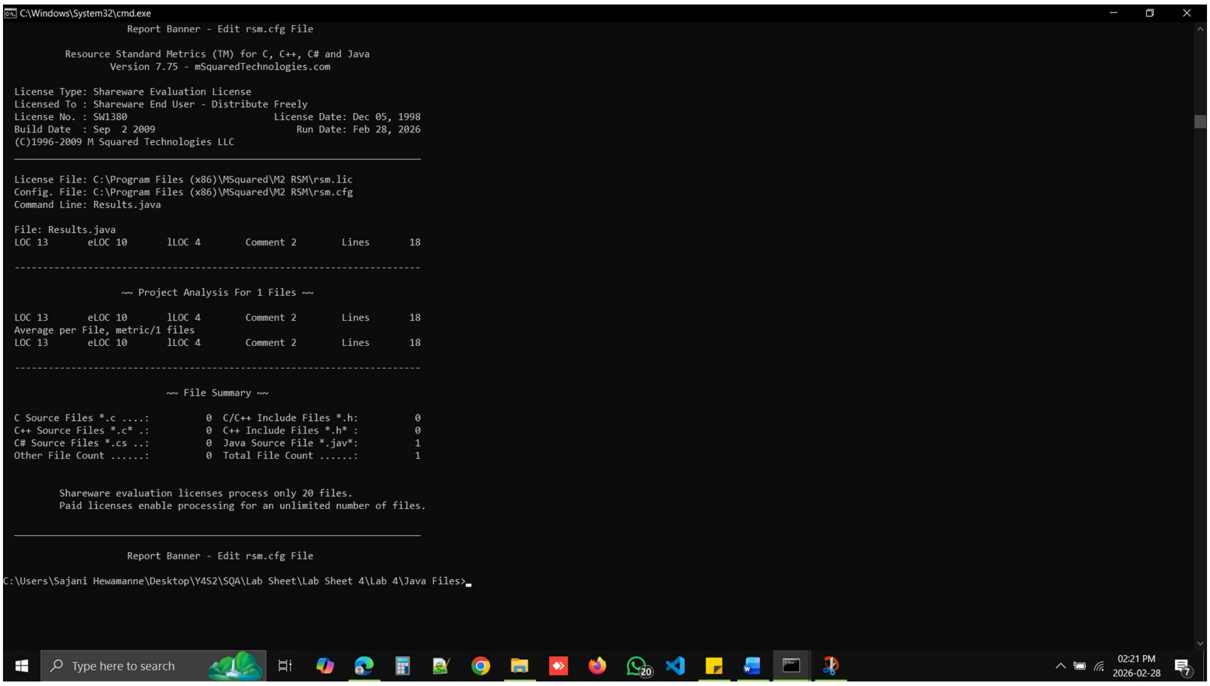
```
C:\Windows\System32\cmd.exe
        -Ta  All Total reports
        -Td  All Files listed in detail
        -Tf  Functions, Classes Namespaces and Notices
        -Ti  Inheritance Tree Only
        -Tl  Summaries only, no listing of functions, classes etc.
        -TL  Language Metric Summary
        -TN  No file names displayed
        -To  Objects Structs, Classes, Interfaces, Namespaces
        -Tp  Quality Profile by notice type
        -Tq  Quality Notice summary list by file
        -Tn  Quality Notices emitted by file and line number
        -Ts  Total summary LOC metrics
        -Tv  Total verbose metrics
        -Tw  Work file metrics differentials only
     -v Verbose metrics for files, LOC, keywords and constructs
     -u User Management and Network License Features
        -us  Show the current user log
        -u"File CFG rsm_user.cfg" User defined RSM configuration
        -u"File XSL rsm_td.xsl" User specified XSL for -X output mode
        -ua  (Network License Only) Archive log/restart
        -ur  (Network License Only) Remove user from current log
     -w Work files for metrics differentials between baselines
        -wb  Show the basis for calculating work productivity
        -w"create top_dir" Create work file, specify baseline top directory
        -wD  Deterministic mode for -wc work file creation
        -wd  Differentials for code lines (Time and Space intensive)
        -we  Do not process equal files between the old and new baselines
        -w"File dat myworkfile.dat"  User specified work file
        -w"File cfg mydiff.cfg"  User defined differential configuration
        -wg  Show the glossary for the report
        -wm  Do not process modified files between old and new baselines
        -wn  Do not process new files between the old and new baselines
        -wp  Show Work Productivity Estimates
        -wr  Do not process removed files between the old and new baselines
        -ws  Show Code Differential Details for each file, line no & Type
        -wv  Verbose mode for metrics differential files
        -wW  Deprecated as of version 6.52, use -wx
        -w"x OlderWork.dat NewerWork.dat" Extract work differentials
     -X XML Output Mode
        -X -T*  All T report switches, reference -T
        -X -wx  Work file differentials extraction
     -x Extended options
        -xNOCONFIG  No configuration file reporting in output
        -xNOCOMMAND No command line reporting in output
        -xUDQNSHOW  Show the User Defined Quality Notices
        -xNOWRAP    Do not wrap long names in reports
     -y Display configuration parameters, export rsm.cfg

C:\Users\Sajani Hewamanne\Desktop\Y4S2\SQA\Lab Sheet\Lab Sheet 4\Lab 4\Java Files>
```

## Get the source code details of the Student.java file on to the screen.

rsm NameOfFile

rsm Results.java

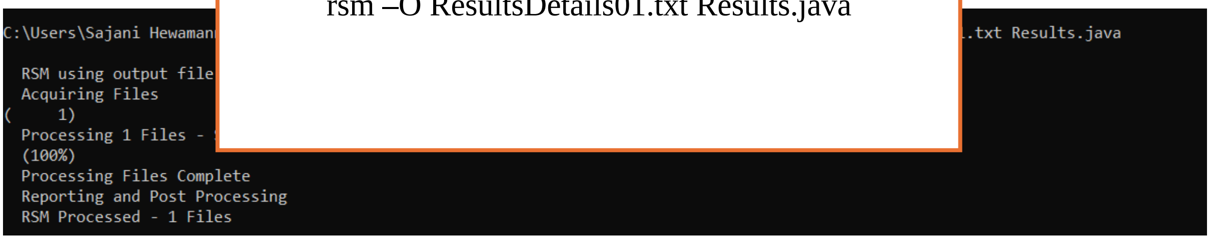
```
C:\Users\Sajani Hewamanne\Desktop\Y4S2\SQA\Lab Sheet\Lab Sheet 4\Lab 4\Java Files>rsm Results.java
```

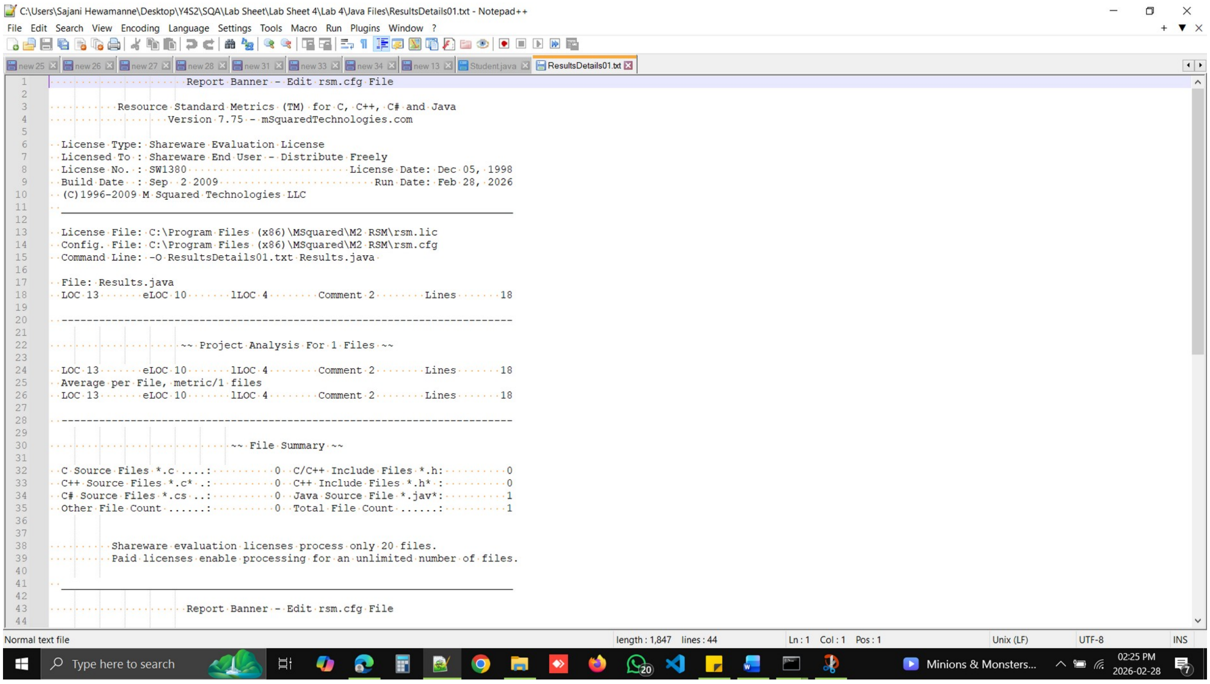



## Get the source code details of the Student.java file to a text file.

rsm –O OutputFileName.txt NameOfFile

rsm –O ResultsDetails01.txt Results.java



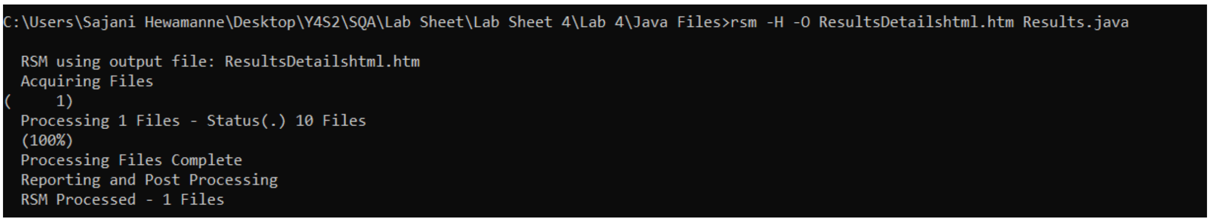


```
C:\Users\Sajani Hewamanne\Desktop\Y4S2\SQA\Lab Sheet\Lab Sheet 4\Lab 4\Java Files\ResultsDetails01.txt - Notepad++

                       Report Banner - Edit rsm.cfg File

           Resource Standard Metrics (TM) for C, C++, C# and Java
                   Version 7.75 - mSquaredTechnologies.com

  License Type: Shareware Evaluation License
  Licensed To : Shareware End User - Distribute Freely
  License No. : SW1380                         License Date: Dec 05, 1998
  Build Date  : Sep  2 2009                        Run Date: Feb 28, 2026
  (C)1996-2009 M Squared Technologies LLC
  ________________________________________________________________________

  License File: C:\Program Files (x86)\MSquared\M2 RSM\rsm.lic
  Config. File: C:\Program Files (x86)\MSquared\M2 RSM\rsm.cfg
  Command Line: -O ResultsDetails01.txt Results.java

  File: Results.java
  LOC 13       eLOC 10       lLOC 4        Comment 2        Lines       18

  ------------------------------------------------------------------------

                      ~~ Project Analysis For 1 Files ~~

  LOC 13       eLOC 10       lLOC 4        Comment 2        Lines       18
  Average per File, metric/1 files
  LOC 13       eLOC 10       lLOC 4        Comment 2        Lines       18

  ------------------------------------------------------------------------

                              ~~ File Summary ~~

  C Source Files *.c ....:           0  C/C++ Include Files *.h:          0
  C++ Source Files *.c* .:           0  C++ Include Files *.h* :          0
  C# Source Files *.cs ..:           0  Java Source File *.jav*:          1
  Other File Count ......:           0  Total File Count ......:          1


          Shareware evaluation licenses process only 20 files.
          Paid licenses enable processing for an unlimited number of files.

  ________________________________________________________________________

                       Report Banner - Edit rsm.cfg File
```

## Get the source code details of the Student.java file to a HTML file.

rsm –H –O OutputFileName.htm NameOfFile

rsm –H –O ResultsDetailshtml.htm Results.java



```
C:\Users\Sajani Hewamanne\Desktop\Y4S2\SQA\Lab Sheet\Lab Sheet 4\Lab 4\Java Files>rsm -H -O ResultsDetailshtml.htm Results.java

  RSM using output file: ResultsDetailshtml.htm
  Acquiring Files
(     1)
  Processing 1 Files - Status(.) 10 Files
  (100%)
  Processing Files Complete
  Reporting and Post Processing
  RSM Processed - 1 Files
```


## What is the purpose of the option – Tf?

**Tf -** Functions, Classes Namespaces and Notices

## Complexity (Param, Return, Cyclo vg, Total)

**Cyclo vg - Cyclomatic complexity - Number of linearly independent paths - Decision statements**

**Param = Parameter Count**

**Return = Number of return count**

**Cyclo vg = Cyclomatic complexity**

Total = Param+ Return+ Cyclo vg

= 6 + 4 +1

= 11

## LOC (eLOC, lLoc, Comment, Lines)

**Function Points**

*(53) is a standard value*

FP(LOC) = Function LOC

= 13/53

= 0.25

FP(eLOC) = Function eLOC

= 10/53

= 0.19

FP(ILOC) = Function ILOC

= 4/53

= 0.08

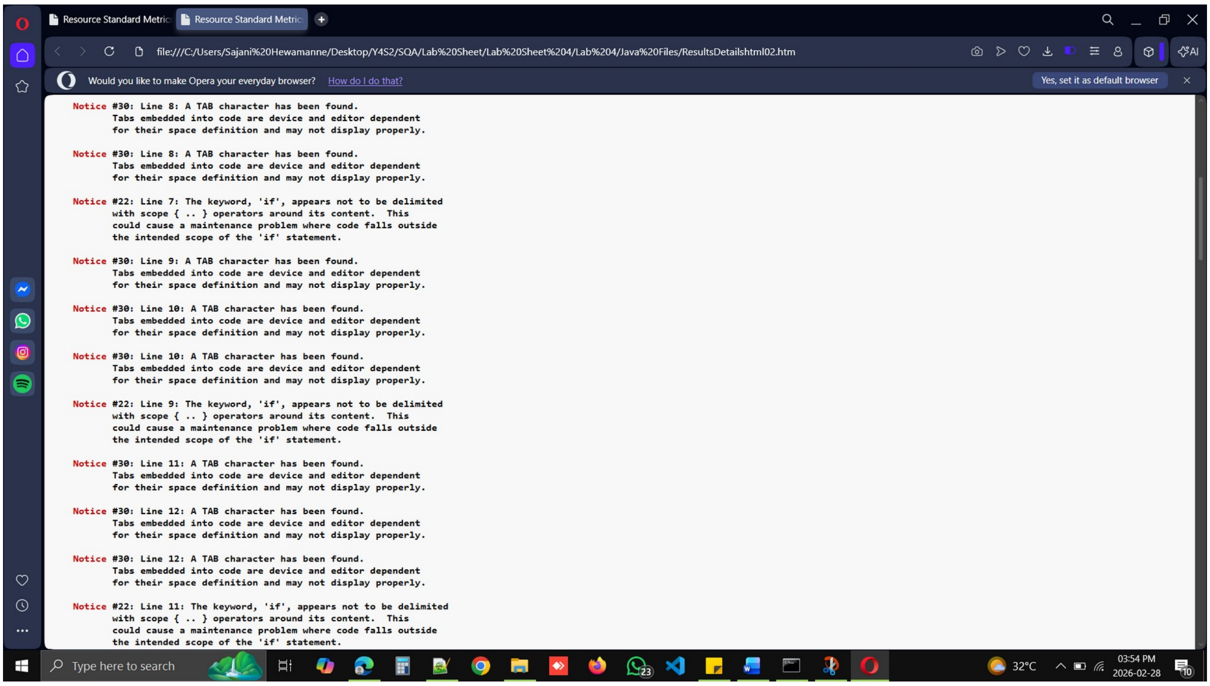
## Get quality notices for the Results.java file.

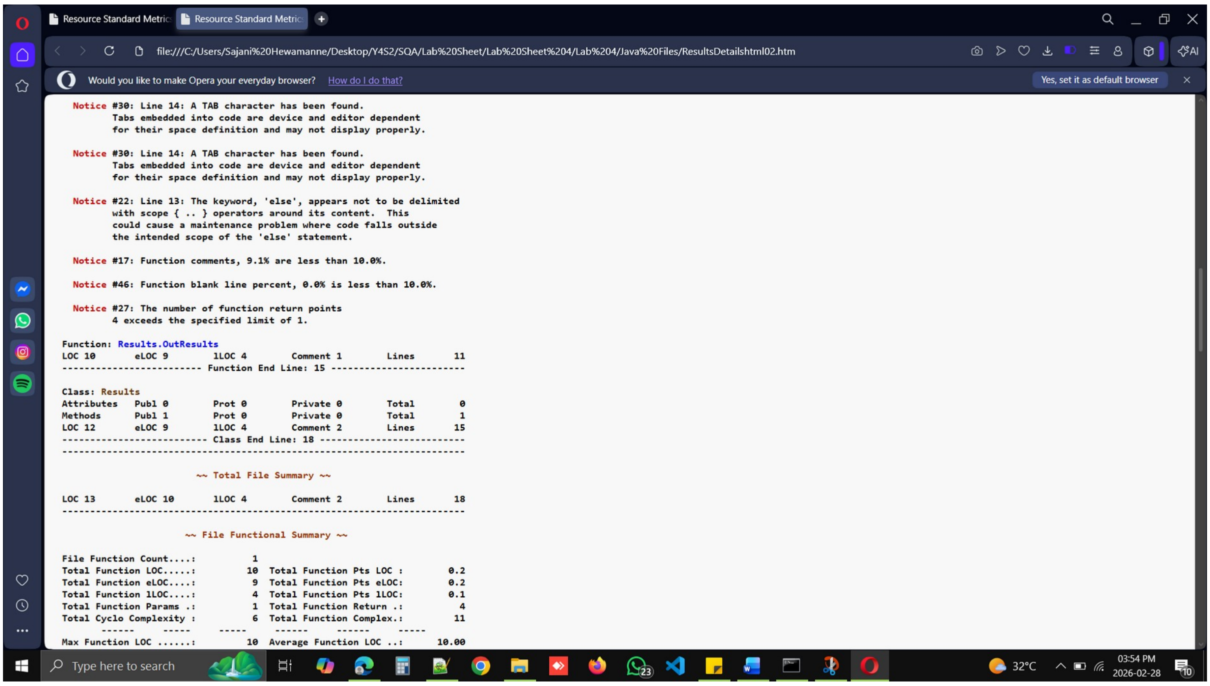
**-fa---->Functional Identification Analysis**

**-o ----> Object metrics classes and structs**

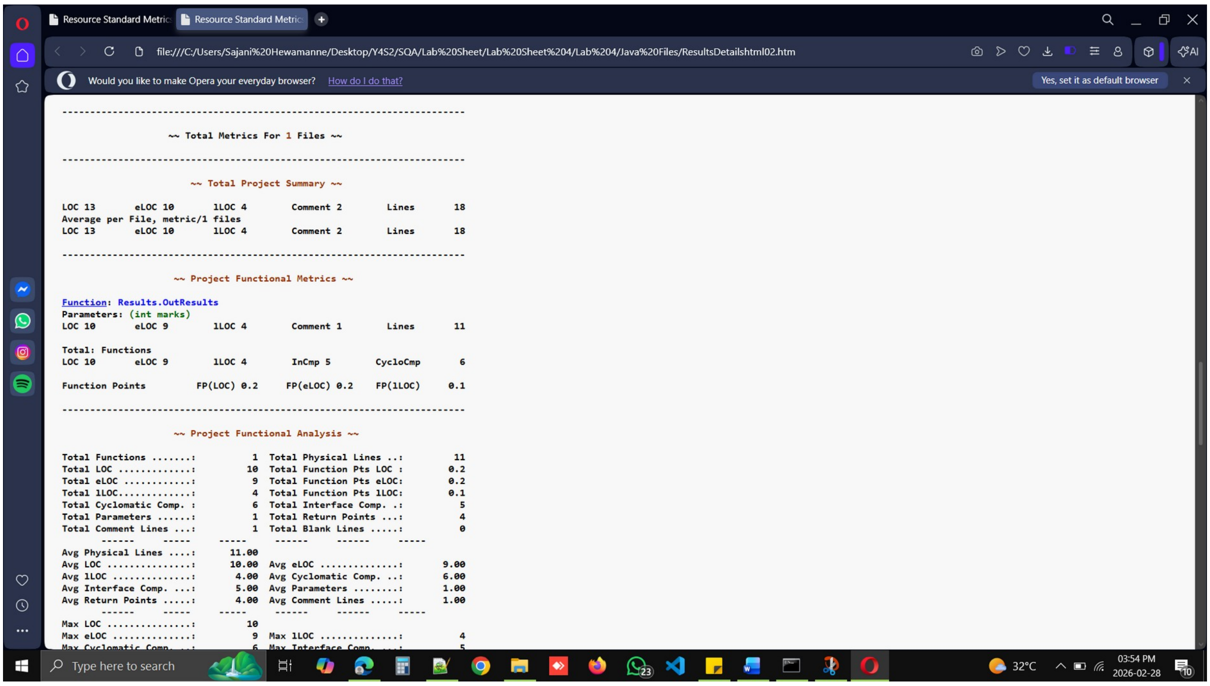
```
rsm -H -O report1.htm -fa -o -n Results.java

rsm -H -O ResultsDetailshtml02.htm -fa -o -n Results.java
```

```
C:\Users\Sajani Hewamanne\Desktop\Y4S2\SQA\Lab Sheet\Lab Sheet 4\Lab 4\Java Files>rsm -H -O ResultsDetailshtml02.htm -fa -o -n Results.java

 RSM using output file: ResultsDetailshtml02.htm
 Acquiring Files
(     1)
 Processing 1 Files - Status(.) 10 Files
 (100%)
 Processing Files Complete
 Reporting and Post Processing
 RSM Processed - 1 Files
```

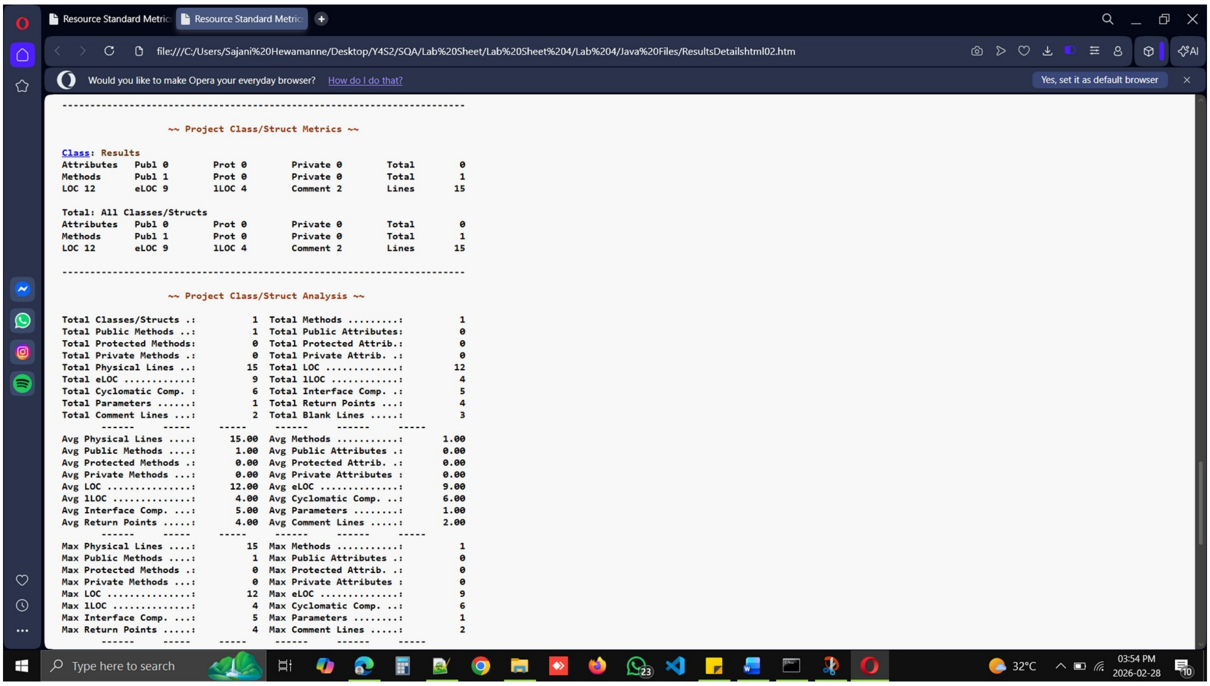





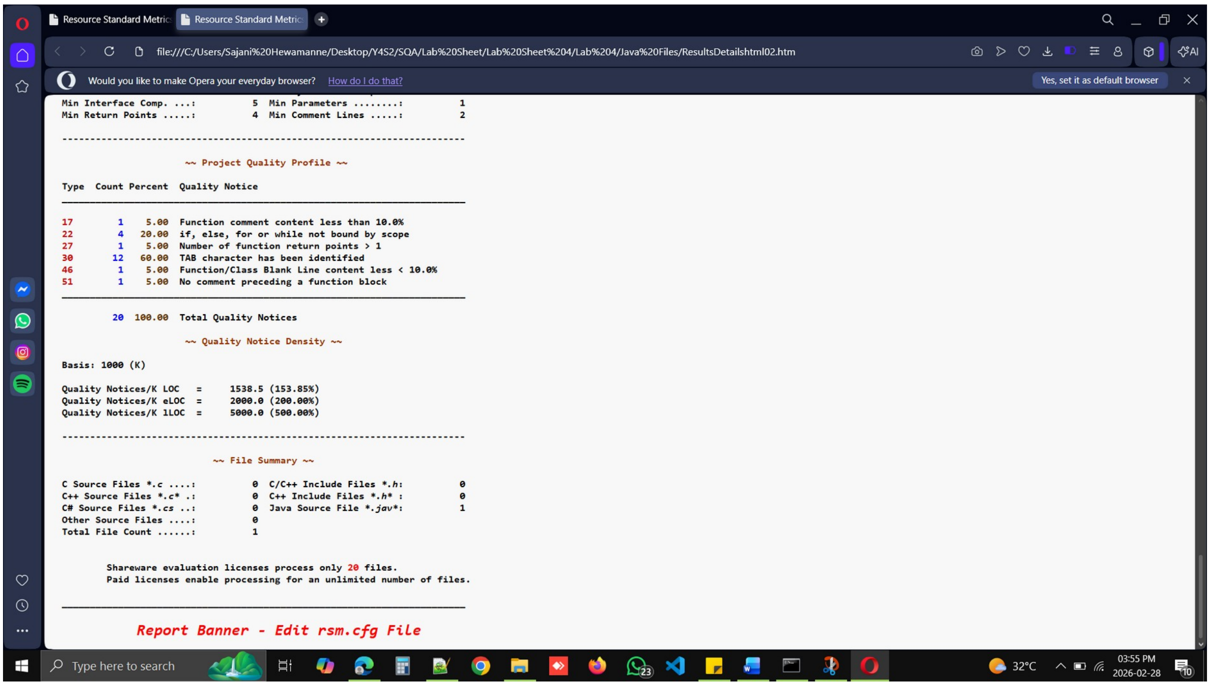
```
Notice #30: Line 14: A TAB character has been found.
       Tabs embedded into code are device and editor dependent
       for their space definition and may not display properly.

Notice #30: Line 14: A TAB character has been found.
       Tabs embedded into code are device and editor dependent
       for their space definition and may not display properly.

Notice #22: Line 13: The keyword, 'else', appears not to be delimited
       with scope { .. } operators around its content.  This
       could cause a maintenance problem where code falls outside
       the intended scope of the 'else' statement.

Notice #17: Function comments, 9.1% are less than 10.0%.

Notice #46: Function blank line percent, 0.0% is less than 10.0%.

Notice #27: The number of function return points
       4 exceeds the specified limit of 1.

Function: Results.OutResults
LOC 10       eLOC 9       lLOC 4       Comment 1       Lines      11
------------------------ Function End Line: 15 ------------------------

Class: Results
Attributes   Publ 0       Prot 0       Private 0       Total       0
Methods      Publ 1       Prot 0       Private 0       Total       1
LOC 12       eLOC 9       lLOC 4       Comment 2       Lines      15
------------------------- Class End Line: 18 -------------------------
-----------------------------------------------------------------------

                        ~~ Total File Summary ~~

LOC 13       eLOC 10      lLOC 4       Comment 2       Lines      18
-----------------------------------------------------------------------

                      ~~ File Functional Summary ~~

File Function Count....:          1
Total Function LOC.....:         10  Total Function Pts LOC :      0.2
Total Function eLOC....:          9  Total Function Pts eLOC:      0.2
Total Function lLOC....:          4  Total Function Pts lLOC:      0.1
Total Function Params .:          1  Total Function Return .:        4
Total Cyclo Complexity :          6  Total Function Complex.:       11
       ------     -----    -----    ------     ------     -----
Max Function LOC ......:         10  Average Function LOC ..:    10.00
```



```
-----------------------------------------------------------------------

                   ~~ Total Metrics For 1 Files ~~

-----------------------------------------------------------------------

                      ~~ Total Project Summary ~~

LOC 13       eLOC 10      lLOC 4       Comment 2       Lines      18
Average per File, metric/1 files
LOC 13       eLOC 10      lLOC 4       Comment 2       Lines      18

-----------------------------------------------------------------------

                    ~~ Project Functional Metrics ~~

Function: Results.OutResults
Parameters: (int marks)
LOC 10       eLOC 9       lLOC 4       Comment 1       Lines      11

Total: Functions
LOC 10       eLOC 9       lLOC 4       InCmp 5       CycloCmp      6

Function Points        FP(LOC) 0.2    FP(eLOC) 0.2   FP(lLOC)     0.1

-----------------------------------------------------------------------

                    ~~ Project Functional Analysis ~~

Total Functions .......:          1  Total Physical Lines ..:       11
Total LOC .............:         10  Total Function Pts LOC :      0.2
Total eLOC ............:          9  Total Function Pts eLOC:      0.2
Total lLOC.............:          4  Total Function Pts lLOC:      0.1
Total Cyclomatic Comp. :          6  Total Interface Comp. .:        5
Total Parameters ......:          1  Total Return Points ...:        4
Total Comment Lines ...:          1  Total Blank Lines .....:        0
       ------     -----    -----    ------     ------     -----
Avg Physical Lines ....:      11.00
Avg LOC ...............:      10.00  Avg eLOC ..............:     9.00
Avg lLOC ..............:       4.00  Avg Cyclomatic Comp. ..:     6.00
Avg Interface Comp. ...:       5.00  Avg Parameters ........:     1.00
Avg Return Points .....:       4.00  Avg Comment Lines .....:     1.00
       ------     -----    -----    ------     ------     -----
Max LOC ...............:         10
Max eLOC ..............:          9  Max lLOC ..............:        4
Max Cyclomatic Comp. ..:          6  Max Interface Comp. ...:        5
```



```
-------------------------------------------------------------------------

                  ~~ Project Class/Struct Metrics ~~

Class: Results
Attributes    Publ 0        Prot 0        Private 0       Total        0
Methods       Publ 1        Prot 0        Private 0       Total        1
LOC 12        eLOC 9        lLOC 4        Comment 2       Lines       15

Total: All Classes/Structs
Attributes    Publ 0        Prot 0        Private 0       Total        0
Methods       Publ 1        Prot 0        Private 0       Total        1
LOC 12        eLOC 9        lLOC 4        Comment 2       Lines       15

-------------------------------------------------------------------------

                  ~~ Project Class/Struct Analysis ~~

Total Classes/Structs .:          1  Total Methods .........:          1
Total Public Methods ..:          1  Total Public Attributes:          0
Total Protected Methods:          0  Total Protected Attrib.:          0
Total Private Methods .:          0  Total Private Attrib. .:          0
Total Physical Lines ..:         15  Total LOC .............:         12
Total eLOC ............:          9  Total lLOC ............:          4
Total Cyclomatic Comp. :          6  Total Interface Comp. .:          5
Total Parameters ......:          1  Total Return Points ...:          4
Total Comment Lines ...:          2  Total Blank Lines .....:          3
       ------     -----     -----     ------     ------     -----
Avg Physical Lines ....:      15.00  Avg Methods ...........:       1.00
Avg Public Methods ....:       1.00  Avg Public Attributes .:       0.00
Avg Protected Methods .:       0.00  Avg Protected Attrib. .:       0.00
Avg Private Methods ...:       0.00  Avg Private Attributes :       0.00
Avg LOC ...............:      12.00  Avg eLOC ..............:       9.00
Avg lLOC ..............:       4.00  Avg Cyclomatic Comp. ..:       6.00
Avg Interface Comp. ...:       5.00  Avg Parameters ........:       1.00
Avg Return Points .....:       4.00  Avg Comment Lines .....:       2.00
       ------     -----     -----     ------     ------     -----
Max Physical Lines ....:         15  Max Methods ...........:          1
Max Public Methods ....:          1  Max Public Attributes .:          0
Max Protected Methods .:          0  Max Protected Attrib. .:          0
Max Private Methods ...:          0  Max Private Attributes :          0
Max LOC ...............:         12  Max eLOC ..............:          9
Max lLOC ..............:          4  Max Cyclomatic Comp. ..:          6
Max Interface Comp. ...:          5  Max Parameters ........:          1
Max Return Points .....:          4  Max Comment Lines .....:          2
       ------     -----     -----     ------     ------     -----
```



```
Min Interface Comp. ...:          5  Min Parameters ........:          1
Min Return Points .....:          4  Min Comment Lines .....:          2

-------------------------------------------------------------------------

                  ~~ Project Quality Profile ~~

Type  Count Percent  Quality Notice
_________________________________________________________________________

17        1    5.00  Function comment content less than 10.0%
22        4   20.00  if, else, for or while not bound by scope
27        1    5.00  Number of function return points > 1
30       12   60.00  TAB character has been identified
46        1    5.00  Function/Class Blank Line content less < 10.0%
51        1    5.00  No comment preceding a function block
_________________________________________________________________________

         20  100.00  Total Quality Notices

                  ~~ Quality Notice Density ~~

Basis: 1000 (K)

Quality Notices/K LOC   =     1538.5 (153.85%)
Quality Notices/K eLOC  =     2000.0 (200.00%)
Quality Notices/K lLOC  =     5000.0 (500.00%)

-------------------------------------------------------------------------

                  ~~ File Summary ~~

C Source Files *.c ....:          0  C/C++ Include Files *.h:          0
C++ Source Files *.c* .:          0  C++ Include Files *.h* :          0
C# Source Files *.cs ..:          0  Java Source File *.jav*:          1
Other Source Files ....:          0
Total File Count ......:          1

        Shareware evaluation licenses process only 20 files.
        Paid licenses enable processing for an unlimited number of files.

_________________________________________________________________________

             Report Banner - Edit rsm.cfg File
```

## List three usages of the RSM.

1. Requirement Documentation

RSM is employed to properly document the entire functional and non-functional needs of the system, ensuring that everyone understands what needs to be done.

2. Communication Between Stakeholders

It serves as a reference tool that facilitates communication among clients, developers, designers, and testers concerning the system’s needs without any form of confusion.

3. Basis for Testing and Validation

RSM is employed as a guideline to validate the system by ensuring that the software developed meets the needs that were identified.

--End of the Report--